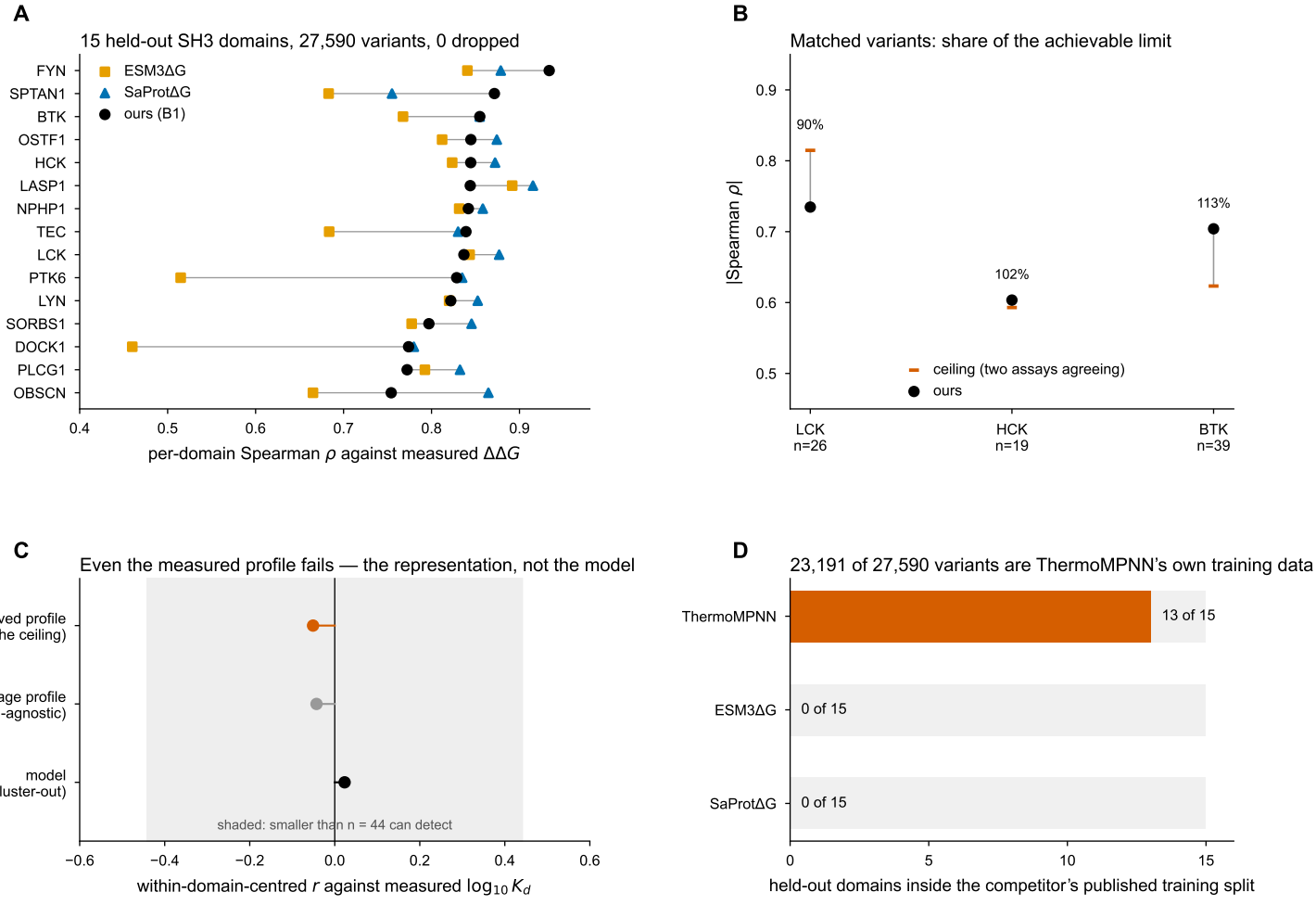
A
15 held-out SH3 domains, 27,590 variants, 0 dropped
ESM3ΔG
SaProtΔG
ours (B1)
FYN
SPTAN1
BTK
OSTF1
HCK
LASP1
NPHP1
TEC
LCK
PTK6
LYN
SORBS1
DOCK1
PLCG1
OBSCN
0.4
0.5
0.6
0.7
0.8
0.9
per-domain Spearman ρ against measured ΔΔG
B
Matched variants: share of the achievable limit
|Spearman ρ|
0.9
0.8
0.7
0.6
0.5
90%
102%
113%
ceiling (two assays agreeing)
ours
LCK
n=26
HCK
n=19
BTK
n=39
C
Even the measured profile fails — the representation, not the model
observed profile
(the ceiling)
average profile
(domain-agnostic)
model
(leave-cluster-out)
shaded: smaller than n = 44 can detect
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
within-domain-centred r against measured log10 Kd
D
23,191 of 27,590 variants are ThermoMPNN's own training data
ThermoMPNN
13 of 15
ESM3ΔG
0 of 15
SaProtΔG
0 of 15
0
5
10
15
held-out domains inside the competitor's published training split